

User agent

MAVRA request handling

1-4 use path

5-7 execution path

Text request



Return amount
in May?

lookup "return amount"

1 Match name

name or alias contains it

store return amount	✓
store return rate	—
store revenue	—

2 Compare versions

did its tables change?

	last check	now
store_returns	14	15
date_dim	3	3

changed: recheck needed

3 Reuse or check

check results are kept per table version

one row per return	reused (from return rate)	✓
one date per date key	kept: date_dim unchanged	✓
every return has a date	not kept: check now	✓

4 Return

only valid definitions

revision 2, valid
store_returns
SUM(sr_return_amt)
sr_status='completed'

the agent writes its SQL from these fields

SQL request

```
SELECT SUM(sr_return_amt)
FROM store_returns
JOIN date_dim ON ...
WHERE sr_status = 'completed'
AND d_moy = 5 AND ...
```

declares: return amount, revision 2

5 Check the declaration

can revision 2 be used?

- ✓ exists, not invalidated
- ✓ still the latest revision
- ✓ SQL keeps the status filter
- ✗ **revision 1: outdated, rejected**

6 Same-snapshot validation

conditions hold on the query's snapshot

a write committed: store_returns now at 16

one row per return	checked in the snapshot	✓
one date per date key	reused: date_dim same	✓
every return has a date	checked in the snapshot	✓

7 Execute

same transaction, same snapshot

run the query on the snapshot
answer **3,294,349.93**
a failed condition rejects the query
and sends the definition to maintenance